



SIDUS

User Manual

SS250mkII/SS250mkIII/SS250mkIV Series Pan & Tilt and SS350mkII/SS350mkIII/SS350mkIV Series Rotator

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**This manual covers the
SS250mkII Series & SS250mkIII & SS250mkIV series products**

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IN THIS MANUAL

This manual contains operation and maintenance information for Sidus SS250mkII/mkIII/mkIV positioners manufactured by Sidus Solutions, Inc, P.O. Box 420689, San Diego, CA, 92142.

This manual is not intended to be a textbook on electricity or electronics. Its primary purpose is to provide information and instructions to experienced operators, electricians, and mechanics that have never operated this equipment. It is the intent of this manual to guide and assist operators and maintenance people in the proper use and care of the equipment.

Use of the manual should not be put off until trouble or need for help develops. Read the instructions before starting the unit. Learn to use the manual and to locate information contained herein. The Appendix is the last section. It contains a list of available options that may be purchased with the unit.

If you have any questions concerning your Sidus product, immediately contact our Service Department by e-mail, mail, telephone or FAX.

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Getting Started

1. Introduction

1.1. Description

Sidus Solutions, LLC of San Diego, CA manufactures the SS250mkII/mkIII/mkIV series Pan & Tilt and SS350mkII/mkIII/mkIV series Rotator underwater positioning units covered by this manual. These products are highly accurate robotic positioning devices which allow precise positioning of underwater cameras, sensors and instrumentation.

Precise motion control is achieved through the integration of stepper motors, low backlash harmonic drive gears and precision bearings. These components are configured in a unique modular design which allows for simplified field service and modification.

The drive gears develop up to 65 foot-pounds of torque and are capable of providing continuous 360-degree rotation in both axes. Power conditioning and control interface circuitry are provided by on-board programmable electronics. All components are enclosed in a pressure rated housing designed to minimize the effects of corrosion in harsh environments.

One of the key features of SS250mkII/mkIII/mkIV series is its fast response time. Once power is supplied to the unit, the start-up message is displayed on terminal window within one second. After the start-up message, the response time for each command sent is typically 150 μ seconds.

In addition to these basic features, Sidus Solutions offers a wide range of options including Pressure Balanced Oil Filled (PBOF) units for deep-water applications, selectable rotation limits, positioning feedback voltages for critical position information, input voltage options and in-house engineering services to meet custom application requirements.



Fig:1SS350mkII Rotator (Aluminum housing)



Fig:2SS250mkII Rotator (Aluminum housing)

1.2. Safety

WARNING: ELECTRIC SHOCK CAN BE FATAL. Do not touch live electrical parts. ELECTRIC ARC FLASH can injure eyes, burn skin, cause equipment damage and ignite combustible material. DO NOT use power cables to break load and prevent tools from causing short circuits. IMPROPER PHASE CONNECTION, PARALLELING, OR USE can damage this and attached equipment.

General

Read, understand and follow all instructions in the Operating/Instruction Manual before installing, operating or servicing the equipment. Keep the manual available for future use by all operators.

Equipment that supplies electrical power can cause serious injury or death or damage to other equipment or property. The operator must strictly observe all safety rules and take precautionary actions. Safe practices have been developed from past experience in the use of power source equipment while certain practices below apply only to electrically powered units.

Shock Prevention

Bare conductors, terminals in the output circuit, or ungrounded electrically live unit can fatally shock a person. Have a certified electrician verify that the unit is adequately grounded and learn what terminals and parts are electrically active. Avoid hot spots on machine. Use proper safety clothing, procedures and test equipment. The electrical resistance of the body is decreased when wet permitting dangerous currents to flow through it. Do not work in damp or wet areas when inspecting or servicing equipment. Stand on a dry rubber mat or dry wood and use insulating gloves when dampness or sweat cannot be avoided. Keep clothing dry and never work alone.

Installation and Grounding of Electrically Powered Equipment.

This equipment must be installed and maintained in accordance with the National Electrical Code ANSI/NFPA 70 or with all applicable codes. A power disconnect switch or circuit breaker must be located at the equipment site. Check the manual, and if applicable, nameplate for voltage, frequency, and phase requirements. If only 3-phase power is available, connect any single-phase rated equipment to only two wires of the 3-phase line. DO NOT CONNECT the equipment grounding conductor (lead) to the third live wire of the 3-phase line as this makes the equipment frame electrically HOT which can cause a fatal shock.

Always connect the grounding lead, if supplied in a power line cable, to the grounded switch box or building ground. If not provided use a separate grounding lead. Ensure that the current (amperage) capacity of the grounding lead will be adequate for the worst fault current situation. Refer to the National Electrical Code ANSI/NFPA 70 for details. Do not remove plug ground prongs. Use correctly mating receptacles.

Output Cables and Terminals

Inspect cables frequently for damage to the insulation and the connectors. Replace or repair cracked or worn cables immediately. Do not overload cables. Do not touch output terminal while equipment is energized.

Service and Maintenance

This equipment must be maintained in good electrical condition to avoid hazards stemming from disrepair. Report any equipment defect or safety hazard to the supervisor and discontinue

use of the equipment until its safety has been assured. Repairs should be made by qualified personnel only. Before inspecting or servicing this equipment take the following precautions:

- a.** Shut off all power at the disconnecting switch or line breaker before inspecting or servicing the equipment.
- b.** Lock switches OPEN (or remove line fuses) so that power cannot be turned on accidentally.
- c.** Disconnect power to equipment.
- d.** If troubleshooting must be done with the unit energized, have another person present who is trained in turning off the equipment and providing or calling for first aid.
- e.** Use of a Ground Fault Interrupt device is strongly recommended to prevent damage to personnel and equipment. Never operate the unit if a hazardous condition is present or suspect.

Fire and Explosion Prevention

This product is for safe area use only and is not to be installed where flammable or explosive material or gases are present. Fire and explosion are caused by electrical short circuits, combustible material near the equipment or unsafe operating conditions. Overloaded or shorted equipment can become hot enough to cause fires by self-destruction or by causing nearby combustibles to ignite. For electrically powered equipment, provide primary input protection to remove short circuited or heavily overloaded equipment from the line.

Bodily Injury Prevention

Serious injury can result from contact with live circuit components inside this equipment.

SHUT DOWN and UNPLUG this equipment for inspection and routine maintenance. Use extreme care in doing necessary troubleshooting and adjustment when equipment is in operation.

1.3. Warranty Statement

1. Sidus Solutions, Inc (hereinafter called SSI) warrants that each new and unused Camera, Pan & Tilt, Light or Controller, (hereinafter called the PRODUCT) is of good workmanship and is free from mechanical defects, provided that (1) the PRODUCT is installed and operated in accordance with the printed instructions of Sidus Solutions Inc. (2) the PRODUCT is used under the normal operating conditions for which it is designed, (3) the PRODUCT is not subjected to misuse, negligence or accident, and (4) the PRODUCT receives proper care, lubrication, protection and maintenance under the supervision of trained personnel.
2. This warranty expires 12 months after shipment by SSI to the first user and is not transferable.
3. This warranty does not apply to primary and secondary switch contacts, cable connectors, carbon brushes, fuses, bulbs and filters unless found to be defective prior to use.
4. SSI DOES NOT WARRANT THE FOLLOWING COMPONENTS: Connectors, cables, camera modules and lenses not modified or assemble by SSI. Many of the foregoing components are warranted directly by the manufacturer to the first user and serviced by a worldwide network of distributors and others authorized to handle claims for component manufacturers. A first user's claim should be presented directly to SSI or such an authorized component service outlet. In the event any component manufacturer has warranted its component to SSI and will not deal directly with a first user then SSI will cooperate with the first user in the presentation of a claim to such manufacturer. Under no circumstances does SSI assume any liability for any warranty claim against or warranty work done by or in behalf of any manufacturer of the foregoing components.
5. This warranty is extended by SSI only to the purchaser of new PRODUCTS from SSI or one of its authorized distributors. The PRODUCTS purchased under this warranty are intended for use exclusively by the buyer and his employees and by no other persons and, therefore, there shall be no third party beneficiary to this warranty.
6. A claim of defects in any PRODUCT covered by this warranty is subject to SSI factory inspection and judgment. SSI's liability is limited to repair of any defects found by SSI to exist, or at SSI's option the replacement of the defective product, F.O.B. (freight on board) factory, after the purchaser at its expense has returned the defective product to SSI's shipping place. Replacement and exchange parts will be warranted for the remainder of the original Warranty, or for a period of ninety (90) days, whichever is greater.
7. UNDER NO CIRCUMSTANCES whatsoever shall SSI and its authorized distributors be liable for any special or consequential damages, whether based on lost goodwill, lost resale profits, work stoppage impairment of other goods or otherwise, and whether arising out of breach of any expressed or implied warranty, breach of contract, negligence or otherwise, except only in the case of personal injury as may be required by applicable law.
8. Continued use of the PRODUCT(S) after discovery of a defect VOIDS ALL WARRANTIES.

Installation & Operation

2. Installation & Operation

2.1. Installation

Care should be observed to install the unit out of the way of traffic so as not to create a hazard either to personnel or equipment.

System cabling should be routed and secured so as not to damage it or interfere with Pan & Tilt motion. When mating the control cable connector, use a light coat of silicone spray lubricant on the connector pins. Never mate rubberized connector without silicone spray lubricant as this may result in damage. This will ease mating and minimize damage to the connector's rubber bond during repeated couplings.

Sidus Solutions also offers internal limits setting of $\pm 170^\circ$ on the pan axis and $\pm 90^\circ$ on the tilt axis which prevent the yoke bracket from running into the pan & tilt housing during operation. Other custom hard stop settings are available from the factory. See the supplied configuration sheet and QC report that ships with the unit for details on your unit.

Ensure electrical connections are in accordance with the wiring configurations for your particular unit. See the supplied configuration sheet and QC report that ships with the unit for details on your unit.

2.1.1. Mounting the Base Plate

Sidus Solutions recommends using the optional 907002321-XX mounting plate, pictured below, for secure installation of the unit. This component provides an optimized mounting platform and provides for additional load bearing.

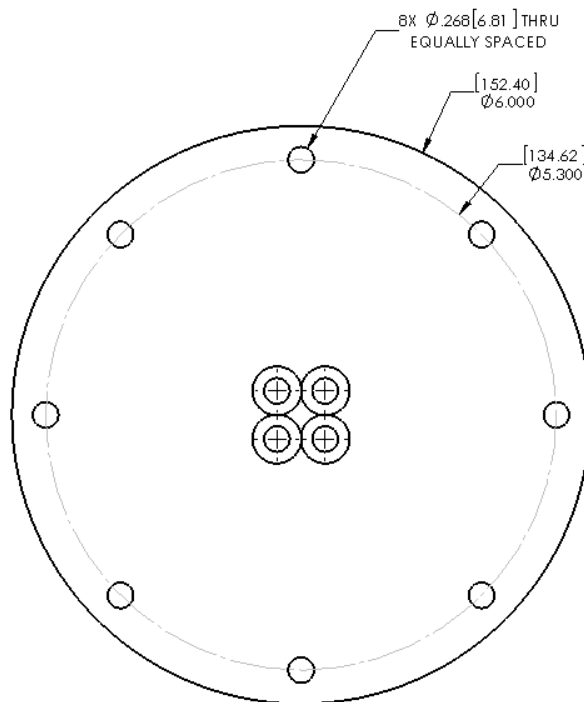


Fig: 3 907002321-01 Base Plate Mounting Dimensions

2.1.2. Direct Shaft Mounting

In the event the SS 907002321-XX is not used there are four 1/4-20 UNC-2B mounting holes on the output shafts for installation purposes. Please refer to the installation drawing below for mounting hole pattern and dimensions and follow all instructions. The SS250mkII/mkIII/mkIV is designed for continuous rotation in both axes.

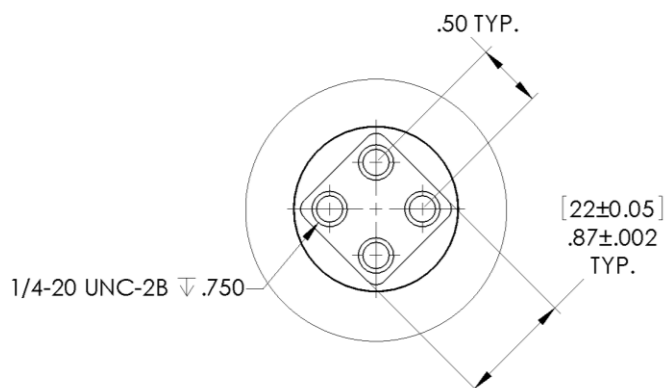


Fig: 4 Drive Shaft Dimensions

2.1.3. Yoke Bracket

Use of the optional yoke-mounting bracket (907001308-01, attached in Figure 3) is recommended for integrating additional items such as cameras, lights and other sensors. Yoke brackets for specific sonars are also available. Check with Sidus to see if your sonar is supported or to inquire about a custom yoke bracket.

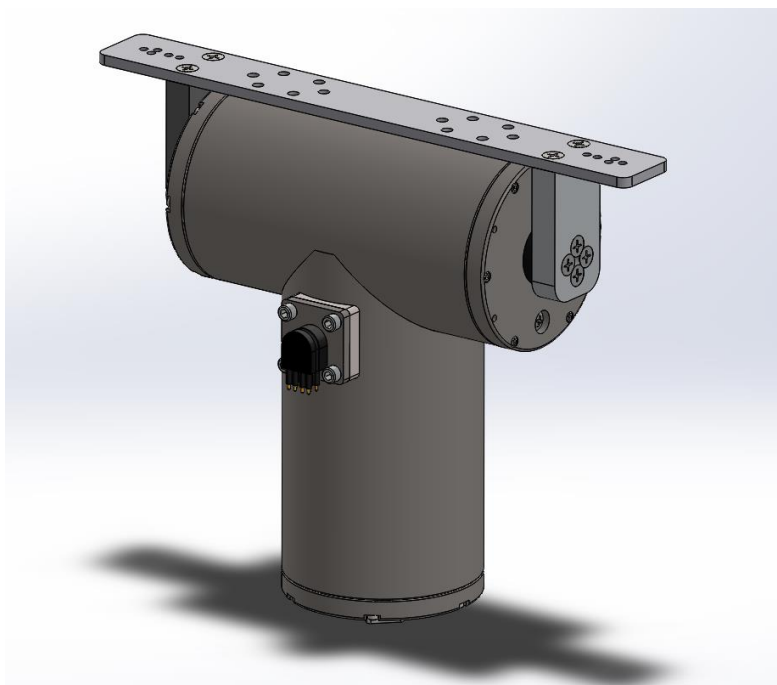


Fig: 5 - SS250mkII with Standard Mounting Bracket

2.2. External Pressure Compensation

The SS250mkII/mkIII/mkIV features an optional port for use with an external oil compensation system. Units with compensation ports are shipped oil filled with a stainless steel plug. Remove the plug using a flat wrench carefully making sure that the port remains at a higher elevation than the rest of the SS250mkII/mkIII/mkIV to prevent oil from spilling out. Install compensation fitting while making sure that connector O-ring firmly seals on the 0.828" diameter sealing surface.

The SS250mkII/mkIII/mkIV is designed for use with Carnation® White Mineral Oil. Additional oil is available through Sidus Service – Part number: FLD WHT OIL. **Compensation with other non-approved oils may damage the internal components and will void warranty.**

Note: it is recommended that all air is purged from the SS250mkII and external compensation system. Internal oil pressure should be a minimum of 1 psi relative to external air pressure.

SIDUS assumes no responsibility for any potential damage or leakage caused by the use of a third party external compensation system.

2.3. Electrical Connection

2.3.1. Connector Pin Out

The connector pinout for SS250mkII/mkIII/mkIV series is shown below:

SS250mkII/mkIII/mkIV series Standard Connector Pinout									
Config	Conn Pinout Name	Pin Number							
		1	2	3	4	5	6	7	8
1	PinOut PT (5-8P) RS485 / RS232	COM	POWER (+24VDC)	SIGNAL GND	D+ / Rx	D- / Tx	NC	NC	NC
3	PinOut PT Analog DC E (8P)	COM	T/CCW	T/CW	P/CW	P/CCW	24vDC	FB Tilt	FB Pan
Additional configurations are available on request. Actual pinout may vary depending on order, please see sales documents to confirm pinout.									

Power Requirement

The power source should be capable of supplying 4 amps continuous. The SS250mkII/mkIII/mkIV units require peak current up to 4 amps will only draw an average of 1 amp. The recommended voltage is 24 VDC, however input supply voltage between 20VDC - 30VDC is acceptable and will not affect performance.

2.3.2. Serial Data Setup

The SS250mkII/mkIII/mkIV units are typically controlled by a computer or PLC system. This requires one serial communication port.

- When the SS250/350 is configured for RS232, only one device per port is supported.
- When configured for RS485, up to 32 devices can be support on a single port.
- The default baud rate is 9600. Data format is 8 data bits, no parity, 1 stop bit ("8,n,1").
- The baud rate can be changed using serial commands listed in section 4 of this manual.

Maintenance & Service

3. Maintenance and Service

3.1. Maintenance

3.1.1. Mounting Hardware

Mounting hardware can vibrate loose and should be regularly inspected to ensure tightness. Mounting hardware that is a part of equipment installed on the unit should also be inspected.

3.1.2. Electrical Connectors

Electrical connectors in dynamic applications are subject to pattern stress. In outdoor and harsh environment applications they are subject to additional thermal and environmental stress and should be inspected bi-weekly for signs of insulation breakdown and fatigue. Ensure mating connectors are clean and free of debris and there are no signs of damage. Ensure the cable is securely routed and has not become tangled or snagged. If damage to the conductor or insulating material is suspect, replace immediately or contact Sidus Solutions for service. See safety section 1.2 of this document.

3.1.3. Housing Care

The Housing has been hard anodized to minimize the effects of corrosion. External mechanical damage is common to equipment in hazardous environments. If the damage is severe base metal may be exposed, corrosion may set in and the structural integrity of the unit may be compromised. Surface finish should be inspected regularly for damage to ensure base metal has not been exposed. If damage is evident please contact Sidus Solutions for evaluation and service options.

3.1.4. O-Ring Seals

O-ring seals are extremely reliable when properly maintained. When kept clean and lubricated they provide years of service. As with all underwater equipment the unit should be cleaned between uses and prior to extended out of water storage. Do this by operating the unit while rinsing in fresh water. This will loosen and remove silt, sand and debris accumulated in the O-ring sealing areas which may later dry, set, and interfere with proper sealing. Using a mild detergent solution and a brush, while not necessary, is a good way of cleaning the unit. Sidus recommends servicing and maintaining the units every 12-18 months to prolong the life of the units.

3.1.5. Pressure Balanced Oil Filled Systems

Pressure Balanced Oil Filled Units (PBOF) should be visually inspected bi-weekly for signs of leakage. The fluid level should also be checked to ensure it is within the normal

range as defined below. The oil level indicator is located in the cavity of the compensator module opposite the output shaft on the end of the tilt axis.

Under normal operating conditions at room temperature the indicator protrudes out of the compensator 0.350-0.450" (without Phillips head screw in pin) to a level near the depth of the cavity. If fluid level drops below this range the indicator will recede into the compensator. Discontinue use of the unit if the indicator pin protrudes less than 0.350" at room temperature.

In cold environs, the indicator will protrude considerably less than at room temperature. In hot environs, the indicator will protrude greater than at room temperature. For the two axes SS250mkII Pan & Tilt, see section 3.1.6 for reference pin height levels for various device temperatures.

A simple way to check for leakage is to wash the unit in warm fresh water to remove outside contamination, set the unit on an oil absorbent pad and let it sit for a minimum of 4 hours. Any fluid seepage will be visible around the O-ring sealing areas and on the pad. If a leak is suspect discontinue use and contact Sidus Solutions for service details.

3.1.6. Standard PBOF Pin Heights

Volume of Oil - 1200-1300 ml / 73-80 in³

Deg F	Compensator Pin height from bottom of the pocket In Air (in.)		
	Nominal	min	max
50	0.171	0.121	0.221
51	0.181	0.131	0.231
52	0.191	0.141	0.241
53	0.202	0.152	0.252
54	0.212	0.162	0.262
55	0.223	0.173	0.273
56	0.233	0.183	0.283
57	0.244	0.194	0.294
58	0.254	0.204	0.304
59	0.264	0.214	0.314
60	0.275	0.225	0.325
61	0.285	0.235	0.335
62	0.296	0.246	0.346
63	0.306	0.256	0.356
64	0.317	0.267	0.367
65	0.327	0.277	0.377
66	0.337	0.287	0.387
67	0.348	0.298	0.398
68	0.358	0.308	0.408
69	0.369	0.319	0.419
70	0.379	0.329	0.429
71	0.390	0.340	0.440
72	0.400	0.350	0.450
73	0.410	0.360	0.460
74	0.421	0.371	0.471
75	0.431	0.381	0.481
76	0.442	0.392	0.492
77	0.452	0.402	0.502
78	0.463	0.413	0.513
79	0.473	0.423	0.523
80	0.483	0.433	0.533
81	0.494	0.444	0.544
82	0.504	0.454	0.554
83	0.515	0.465	0.565
84	0.525	0.475	0.575
85	0.536	0.486	0.586

Deg F	Compensator Pin height from bottom of the pocket In Air (in.)		
	Nominal	min	max
86	0.546	0.496	0.596
87	0.556	0.506	0.606
88	0.567	0.517	0.617
89	0.577	0.527	0.627
90	0.588	0.538	0.638
91	0.598	0.548	0.648
92	0.609	0.559	0.659
93	0.619	0.569	0.669
94	0.629	0.579	0.679
95	0.640	0.590	0.690
96	0.650	0.600	0.700
97	0.661	0.611	0.711
98	0.671	0.621	0.721
99	0.682	0.632	0.732
100	0.692	0.642	0.742
101	0.702	0.652	0.752
102	0.713	0.663	0.763
103	0.723	0.673	0.773
104	0.734	0.684	0.784

Note:

- 1) Temperature here represents ambient temperature/Oil temperature during installation & degassing process.
- 2) During run test or after, take average of the pan and tilt axis temperature to verify corresponding pin height at that average unit temperature.

** This chart applies to SS250 Pan & Tilt unit

3.1.7 Back EMF Damage to Motor Controller

Under some conditions, the SS250 may be damaged by highly eccentric loads. When power to the SS250 is turned off, a highly eccentric load can cause the tilt axis to sag due to the effect of gravity. This is commonly referred to as back driving the motor. When back driving a motor, the motor acts as a generator and can generate very high voltages that can damage the motor drive electronics. This is true for all motors.

Electric motors are designed to convert electrical power into a mechanical motion (a rotating shaft). Electric generators are exactly the opposite; they convert mechanical energy (a rotating shaft) into electrical power. Electric motors and electric generators are very similar in design and construction. So similar, in fact, that an electric motor can be used as an electric generator and vice versa. If a force is applied to the shaft of a motor to turn it, the motor will act like an electric generator and try to generate electricity. This is called a Back Electromotive Force (Back EMF). The magnitude of the Back EMF depends on how hard and how fast the motor shaft is turned. At low speeds relatively little Back EMF is generated. At higher speeds, large amounts of Back EMF are generated and can damage the electronics used to control the motor.

Some motors, such as BLDC motors, are difficult to back drive when they are not powered because a large residual torque (cogging torque) due to internal magnetic forces in the motor has to be overcome before it will rotate. The cogging torque times the reverse gear ratio of the gearhead can be quite difficult to overcome. Stepper motors are relatively easy to back drive because there is very little internal magnetic force in the motor when not powered and therefore very low static torque, even when multiplied by the reverse gear ratio. When BLDC motors are not rotating, it is customary to turn off power to the motor because there is little risk that it can be back-driven. When a stepper motor is not turning, it is customary to continue to power the motor, usually at a reduced power level, to prevent unintended rotation.

However, when power to the SS250 is turned off, eccentric loads may free-fall due to gravity. Depending upon the amount of torque back driving the motor due to this eccentric load, and the resulting speed of rotation, it may be possible to damage the motor controller electronics.

It is impossible to accurately predict how much Back EMF will result from an eccentric load because it depends upon many variables that are not controlled or predictable and vary significantly over time, temperature and from unit to unit. Friction in seals, the viscosity of the compensation fluid, variations in the efficiency of the gearhead are just some of the things that can affect how fast the tilt axis will freefall. The cogging torque of a stepper motor is never specified by the manufacturer because it is unpredictable and not constant. However, it is always true that the faster a back driven motor rotates the higher the Back EMF will be.

The power generated by a back driven motor can be very high. For example, if an eccentric load causes a torque of 10Nm at a speed of 7.5rpm (90° free fall in 2 seconds) the electrical power generated would be a fairly high percentage of the applied mechanical power:

Mechanical Power (kW) = Torque (N.m) x Speed (RPM) / 9.5488 = 10 x 7.5 / 9.5488 = 7.854kW = 7,854W

This high level of power cannot be easily absorbed by the motor controller and can cause damage to the electronics.

Technical Information

4. Technical Information

4.1. Electrical

Description	Specification
Input Voltage	20-30 VDC
Drive Current	0.3 – 2.0 A per axis
Output Torque	65 ft-lbs (75 Nm)
Maximum Payload	150 lbs (68 kg)
Angular Speed	0.5 to 20 deg/sec
Position Feedback mkII/mkIII	Resolution - 0.0879 deg Accuracy - 0.0879 deg
Position Feedback mkIV/mkIV	Resolution - 0.01 deg Accuracy - 0.01 deg
Control	RS232 or RS485 2400-115.2k baud, 8 bit, no parity, 1 stop bit
Response Time after the start-up message	150 µsecond
Connector	See appendix

4.2. Mechanical

Description	Specification
Housing Material	Aluminum, 6061-T6 Hard Anodized, Stainless Steel Electropolished
Fasteners	316 Stainless Steel
Gears	Precision strain wave gearing Backlash: 0.05°
Dimensions	11.5 in x 9.7 in x 5.1 in (292mm x 246mm x 137mm)
Weight in Air	Al 22.9lbs (10.4kg)/SS 49.8lbs (22.6kg)
Weight in Water	Al 13.5lbs (6.1kg)/SS 29.36lbs (13.2kg)
Pressure Compensation	Internal Diaphragm (standard)

External port (optional)

4.3. Environmental

Description	Specification
Operating Depth	3000 m standard internal compensation 6000 m with external compensation
Operating Temperature	-20°C to +50°C
Storage Temperature	-30°C to +60°C

4.4. Sidus Solutions Protocol

NOTE: This section applies only to the SS250/SS350mkII, mkIII, and mkIV series units with RS485/RS232 serial interface.

Key Features:

- When device is configured for RS232, only one device per port is supported.
- Supports device addressing, up to 52 instances of each of 4 types of equipment.
- All communications in ASCII characters (dumb terminal compatible).
- All commands and responses are same length- 12 bytes.
- All commands and data are echoed by the device being addressed.
- SS250mkII/SS250mkIII/SS250mkIV series unit acts as a slave and never initiates the communication.
- No error correction except for echo from addressed device.
- Only a single command is allowed per command string.
- The default factory setting for RS232 protocol- 9600 baud, 8 data bits, No parity, 1 stop bit or as specified by Factory Acceptance Test.
- The baud rate can be changed using serial command.

4.4.1. General Description

The commands used to control and operate SS250mkII/mkIII/mkIV series are mainly divided into 4 categories. In addition to this, there are 2 special commands for controlling SS250mkII/mkIII/mkIV series devices with other Sidus products such as PAN and TILT, CAMERA, LIGHTS and LASER connected on the same RS485/RS232 port. All "ASCII" commands consist of 12 ASCII characters: 10 character for Sidus commands and 2 character for line terminator (<CR> and <LR>). The command response follows the same format.

Command/ Response Format												
Description	HEADER	ADDR	COMMAND			DATA				TERMINATOR	END COMMAND	
Character Count	1	2	3	4	5	6	7	8	9	10	11	12

Fig.5: Components of Sidus Protocol

The different components of ASCII commands are described below:

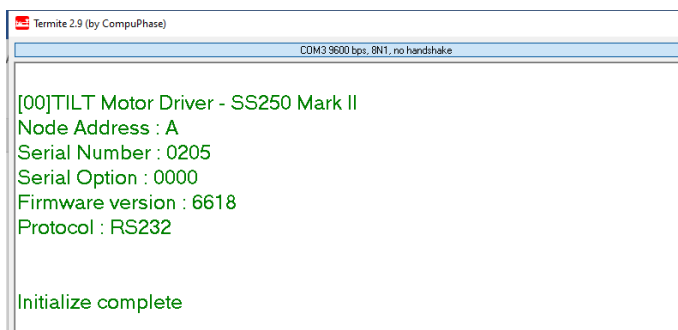
Header: For SS250mkII/mkIII/mkIV devices, the first command byte is the non-alphanumeric character "#" for PAN Axis and "\$" for TILT Axis. Other characters are used for cameras, lights, etc.

Sidus Product	ASCII Header
Light Source	&
Pan Axis	#
Tilt Axis	\$
Camera	%

All Devices	!
-------------	---

Table 1: Header dedicated for various Sidus Products

Node Address: The second command byte is an ASCII character in the range of A-Z or a-z. This byte is used to identify a specific SS250mkII series/SS250mkIII unit. All units must have different “**Node Addresses**”. The node address of the unit for SS250mkII series/SS250mkIII is normally displayed in the power-up/ start-up message. A sample of power up message with a “**Node Address**” of “**A**” for “ONLY” TILT axis is shown below:



```

Termite 2.9 (by CompuPhase)
COM3 9600 bps, 8N1, no handshake

[00]TILT Motor Driver - SS250 Mark II
Node Address : A
Serial Number : 0205
Serial Option : 0000
Firmware version : 6618
Protocol : RS232

Initialize complete

```

Fig.6: Power-up Message of SS250mkII series/SS250mkIII using TERMITE terminal emulator

Command Mnemonic: The next three bytes are command mnemonic. The command mnemonics to control and operate SS250mkII series/SS250mkIII unit is divided into 5 categories and are explained below in detail.

Data Field: The next four bytes are the “**Data Field**”. The four data bytes are sent regardless of how many are required. They are used in combination with “**Command Mnemonic**” and are explained below. All the numbers are positive integers in the range of 0000-9999

Command Terminator: The command terminators are “R” for Read operation or “W” for write operation.

End of Command/ Line Terminator: The next two bytes are ASCII carriage return and line feed characters.

4.4.2. Command Mnemonics for SS250mkII and SS250mkIII series

The command mnemonics for SS250mkII series and SS250mkIII series are divided into 5 categories:

1) Device Configuration

DAD (Device Address): This command is used to configure/modify the node address of the axis. The allowable data field value are: A-Z and a-z. The factory default value is “A”. This command is “Read” and “Write” command.

Example:

Command Sent: **\$ADAD000BW**; Modify the node address “TILT” axis with
node address A to B

Response Received: **\$BDAD000BW**; Acknowledgement received “TILT” axis with node address B.

DAK (Device Acknowledge): This command is used to configure the acknowledgement mode of the axis. The allowable data field value are "0000 for no acknowledgement", "0001 for terse acknowledgment mode" and "0002 for verbose acknowledgement". The factory default setting is "0002" (verbose acknowledgement mode). This command is "Read" and "Write" command.

Example:

Command Sent: **#ADAK0000R**; Read the acknowledgment setting of "PAN" axis with node address "A"

Response Received: **acknowledgement settings: full labels**
#ADAK0002R; "PAN" axis with node address A has the acknowledgement setting as "Verbose"

Command Sent: **#ADAK0001W**; Set the acknowledgment setting of "PAN" axis with node address "A" as Terse response

Response Received: **acknowledgement settings: basic with no labels**
#ADAK0002R; "PAN" axis with node address A has the acknowledgement setting as "Terse"

DBD (Device Baud Rate): This command is used to configure the communication baud rate of the axis. The allowable data field value are: "0024 for 2400bps", "0048 for 4800bps", "0096 for 9600bps", "0192 for 19200bps", "0384 for 38400bps", "0576 for 57600bps", "1152 for 115200bps". The factory default value is 0096 (i.e. 9600 Baud Rate). This is the "Read" and "Write" command.

Example:

Command Sent: **\$ADBD0000R**; Read the Baudrate setting of "TILT" axis with node address "A"

Response Received: **device baud**
\$ADBD0096R; "TILT" axis with node address A has the baud rate as 9600 bdps

GSP (Global Set Protocol): This command is used to modify the communication protocol of the axis from RS232 to RS485 and vice versa on-field or during application. The allowable data field is "0 for RS232" and "1 for RS485". For the axis to successful change the protocol, power-cycle to unit is required. This command is "Write" only command. Sidus is not liable if the axis becomes non-responsive while changing the protocol.

Example:

Command Sent: **\$BGSP0001W**; Switch protocol of "TILT" axis with node address B to RS485

Response Received: **device protocol - cycle power**
\$BGSP0001W; Acknowledgement received from "TILT" axis with node address B; now cycle power

DIV (Device Inversion): This command is useful if the SS250mkII series and SS250mkIII series is attached as "UPSIDE DOWN" means the rotation of both axis is to be "Inverted". This command is used to configure the axis in inverted manner; all the clockwise movement command will perform counter clockwise movement and vice versa. The allowable data field value are "0 for device id not inverted" and "1 for device inverted". This command is "Read" and "Write" command. The factory default is "0" (device not inverted) unless otherwise specified at the time of purchase.

Example:

Command Sent: **\$CDIV0001W**; Invert the "TILT" axis with node address C

Response Received: **device inverted**

\$CDIV0001W; Acknowledgement received from "TILT" axis with node address C

Command Sent: **#ADIV0000R**; Read the "Inversion" setting of "PAN" axis with node address A

Response Received: **device not inverted**

\$ADIV0000R; "PAN" axis with node address A is not inverted

MRA (Motor Report All): This command is used to display the serial number of the axis and start-up/power-up message. This is "Read" command only.

Example:

Command Sent: **\$AMRA0000R**; Read the serial number "PAN" axis with node address A

Response Received: **serial number**

\$AMRA0205R

TILT Motor Driver - SS250 Mark II

Node Address : A

Serial Number : 0205

Serial Option : 0000

Firmware version : 6618

Protocol : RS232

DVG (Device Virgin): This command is used restore the factory default setting of the axis. The allowable data field value is 0000. This command is "Write" command only. The power cycle is required for the axis to configure the factory default setting after the "DVG" command is sent. Sidus is not liable if the axis becomes non-responsive while changing the protocol.

Example:

Command Sent: **#BDVG0000W**; Restore the factory default setting for "PAN" axis with node address B

Response Received: **hardware default values- power cycle device**

MSS (Motor Save Setting) and MRS (Motor Restore Setting): Each axis has 3 user profile in addition to the factory default in order to save the user configuration settings. The command "Motor Save Setting" is used to save user configuration settings and command "Motor Restore Setting" is used to restore the configuration setting into the current user profile. The allowable data field values are "0000 for Factory Default" "0001 for User-Profile-1" "0002 for User-Profile-2" and "0003 for User-Profile-3". It is advisable if the factory default profile is left unmodified. This command is "Write" command only.

Example:

Command Sent: **#AMSS0003W**; Save the current configuration setting of "PAN" axis into User Profile-3

Response Received: **save user settings**

#AMSS0003W; All the current settings are save in user profile-3 for "PAN" axis with node address A

Command Sent: **#AMRS0003W**; Restore the User Profile-3 configuration setting of “PAN” axis into current user profile

Response Received: **restore settings**
#AMSS0003W; Acknowledgement received from “PAN” axis with node address A

2) Motor Movement – Basics

MMF (Motor Move Forward): This command is used to move the axis in the “clockwise direction” at the speed specified in the data field value. The allowable value for data field is 0050 to 2000. The data field value 0050 means 0.5 degrees/sec and 2000 means 20.00 degrees/sec. If the data field is 0000 then axis stops with brakes “on”. This command is “write” only. For this command two responses are received: first response is acknowledge of the command is received and second indicates the current position of axis.

Example

Command Sent: **#AMMF1050W**; Move “PAN” axis with node address A in clockwise direction at 10.5 degrees/sec

Response Received: **move forward**
#AMMF1050W; Acknowledgement of the command received from the “PAN” axis with node address A
#AMRL6024R; “PAN” axis stopped at 6024 count (90 degrees)

*This command can also be used as a braking mechanism if the user wants the brake setting to be applied. In this case, the data field would be set to 0000, to indicate that the motor should not be moving but current should still be applied to the motor.

MMB (Motor Move Backward): This command is used to move the axis in the “counter clockwise direction” at the speed specified in the data field value. The allowable value for data field is 0050 to 2000. The data field value 0050 means 0.5 degrees/sec and 2000 means 20.00 degrees/sec. If the data field is 0000 then axis stops with brakes “on”. This command is “write” only. For this command two responses are received: first response is acknowledge of the command is received and second indicates the current position of axis.

Example:

Command Sent: **#AMMB0800W**; Move “PAN” axis with node address A in counter clockwise direction at 8 degrees/sec

Response Received: **move backward**
#AMMB0800W; Acknowledgement of the command received from the “PAN” axis with node address A
#AMRL3976R; “PAN” axis stopped at 3976 count (-90 degrees)

*This command can also be used as a braking mechanism if the user wants the brake setting to be applied. In this case, the data field would be set to 0000, to indicate that the motor should not be moving but current should still be applied to the motor.

MML (Move Motor Movement): This command moves the axis to the “location value” specified in the data field at speed set by “Motor Speed” command (MSP). The “location value” is decimal count value. For degree to count conversion refer to section “Position Calculation” and quick guide refer to “Degree to Count Conversion Table”. This command is “Write” only. Just like, MMF and MMB command two responses are received from this command: first response is acknowledge of the command is received and second indicates the current position of axis.

Example:

Command Sent: **#AMML6023W**; Move “PAN” axis with node address A to 6023 count (+90 degrees) at speed specified by MSP command

Response Received: **move location**

#AMML6023W; Acknowledgement of the command received from
“PAN” axis with node address A

#AMRL6024R; “PAN” axis with node address A
stopped at 6024 count (+90 degrees)

MRL (Motor Read Location): This command reads the current positional value of the axis. The current positional value is indicated by data field of the response received. This value is decimal count value. For degree to count conversion refer to section “Position Calculation” and quick guide refer to “Degree to Count Conversion Table”. This is “Read” command only.

Example:

Command Sent: **#AMRL0000R**; Read the current location of “PAN” axis with node address A

Response Received: **location**

#AMRL6024R; The current location of “PAN” axis with node address A is
6024 count (+90 degree)

MSP (Motor Speed): This command set the speed value of the axis for Motor Move Location (MML) location. The “location value” is decimal count value. The allowable value for data field is 0050 to 2000. The data field value 0050 means 0.5 degrees/sec and 2000 means 20.00 degrees/sec. If the data field value is 0000 then motor stops with brakes “on”. This is a “Read” and “Write” command.

Example:

Command Sent: **#AMSP0700W**; Set speed of “PAN” axis with node address A as 7 degrees/sec

Response Received: **speed set**

#AMSP0700W; Speed of the “PAN” axis with node address A is set as
7 degrees/sec

Command Sent: **#AMSP0000R**; Read the set speed of “PAN” axis with node address A

Response Received: **speed set**

#AMSP1000R; Speed of “PAN” axis with node address A is 10 degrees/sec

MTQ (Motor Torque): This command sets the “drive current” (in percentage) of axis to the value specified in the data field. The allowable range of data field value is “0000 for 0%” to “0075 for 75%”. The factory default setting is 0033 for 33%. This is a “Read” and “Write” command.

Example:

Command Sent: **\$AMTQ0000R**; Read torque setting for “TILT” axis with node address A

Response Received: torque setting

\$AMTQ0075R; Torque setting of “TILT” axis with node address A is 75%

Command Sent: **\$AMTQ0050W**; Set torque setting for “TILT” axis with node address A as 50%

Response Received: torque setting

\$AMTQ0050W; The torque setting of “TILT” axis with node address A is 50%

***For older firmware versions the allowable limit and factory default maybe be different.**

MBS (Motor Brake Set): This command sets the “drive current” (in percentage) of axis to the value specified in the data field. The allowable range of data field value is “0000 for 0%” to “0100 for 100%”. The factory default setting is “0050 for 50%”. This is a “Read” and “Write” command.

Example:

Command Sent: **#AMBS0075W**; Set brake setting for “PAN” axis with node address A as 75%

Response Received: brake set

#AMBS0075W; Brake setting “PAN” axis with node address A is set to 75%

Command Sent: **#AMBS0000R**; Read brake setting for “PAN” axis with node address A

Response Received: brake set

#AMBS0075R; Brake setting of “PAN” axis with node address A is 75%

MST (Motor Stop): This command is used to stop the movement of the axis. The condition of the “Motor Stop” or the “Status of the Stop” condition is described in the response received. This command is “Write” command only.

If the return data field is:

- 0000: Normal stop operation (motor is command to stop and it is stopping with normal conditions)
- 0100: Motor slipped its counting or possibly stopped because of hard stop.
- 0080: Latch stall detected
- 0040: Motor stops because of stall detection. The increased torque load causes the motor to stall
- 0020: If the excessive current is detected on the output of gate of FET connected to channel B; Channel B pre-drive fault occurs
- 0010: If the excessive current is detected on the output of gate of FET connected to channel A; Channel A pre-drive fault occurs
- 0008: If motor measures under voltage; it stops.
- 0004 Motor measures the over current on channel B; which results in shut down
- 0002: Motor measures the over current on channel A; which results in shut down
- 0001: Motor stops because of the over temperature

Example:

Command Sent: **#AMST0000W**; Stop PAN” axis with node address A

Response Received: **driver stop**

-normal

#AMST0000W; PAN” axis with node address A
stopped under normal condition

*If the user wants the brake setting to be applied when the motor is stopped, the alternative MMF and MMB commands with a data field of 0000 should be used. This command will not take into consideration the current brake setting.

DES (Device Encoder Slip): The encoder misses count because of number of reasons: malfunctioning of encoder or code wheel over a period of time. This command is used to set maximum allowable number of encoder counts the axis can “miss” before axis report “motor slip” error and causes the axis to stop abruptly. The allowable data field value is 0001 for 1 encoder count to 0010 for 10 encoder counts. If the set value is 0000 then encoder slip feature is disabled for the axis. The factory default value is 0008 for 8 encoder count. This command is “Read” and “Write” command.

Example

Command Sent: **\$CDES0000R**; Read “Encoder Slip” setting of “TILT” axis with node address C

Response Received: **device encoder slip detection enabled**

\$CDES0008R; Encoder Slip setting for “TILT” axis with node address C is
8 encoder counts

MAC (Motor Acceleration) [Firmware C1.0 and below]: The “Motor Acceleration” command is used to set the distance from the target location at which motor speed will start to decrease. The default value is 0.

Example:

Command Sent: **#AMAC0000R**; Read the acceleration setting of “PAN” axis with node address A

Response Received: **acceleration rate**

#AMAC0010R; the acceleration setting of “PAN” axis with node address A is 10.

Command Sent: **#AMAC0050W**; Write the acceleration setting of “PAN” axis with node address A

Response Received: **acceleration rate**

#AMAC0050W; the acceleration setting of “PAN” axis with node address A is 50.

MAC (Motor Acceleration) [Firmware D1.1 and above]: This command is used to adjust the speed of the acceleration / deceleration ramp. Having the acceleration ramp enabled improves accuracy when approaching the target location. The allowable range of values for this is 0 to 999. A value of 0 will disable the acceleration ramp. A value of 10 will give a fast acceleration rate while a value of 999 will give the slowest acceleration rate. Too slow of an acceleration and moving small distances may not allow the device to reach full speed. The default acceleration value is 10.

Example:

Command Sent: #AMAC0000R; Read the acceleration setting of "PAN" axis with node address A

Response Received: acceleration rate

#AMAC0010R; the acceleration setting of "PAN" axis with node address A is 10.

Command Sent: #AMAC0050W; Write the acceleration setting of "PAN" axis with node address A

Response Received: acceleration rate

#AMAC0050W; the acceleration setting of "PAN" axis with node address A is 50.

MAP (Motor Approach Distance) [Firmware D1.1 and above]: This command is used to adjust the distance from the target location that the motor switches to the slowest possible speed. This allows high accuracy when approaching the target location. The allowable values are 0 to 9999. A value of 0 disables this feature. The command value represents the distance from the target location that the motor speed is set to 0.5 degrees per second. Each value step represents 0.0879 degrees. The default value is 5.

Example:

Command Sent: #AMAP0000R; read the approach distance setting.

Response Received: approach counter

#AMAP0005R; approach distance settings for "PAN" axis with node address A is 5 count.

Command Sent: #AMAP0010W; write the approach distance setting

Response Received: approach counter

#AMAP0010W; approach distance settings for "PAN" axis with node address A is set to 10 count.

MDB (Motor Dead Band): This command sets the distance from the target location that the device will consider as reaching the target. Each value step represents 0.0879 degrees. The allowable range is 1 to 9999. The default value is 1.

Example:

Command Sent: #AMDB0000R; read the dead band setting.

Response Received: move to dead band

#AMDB0001R; Dead band settings for "PAN" axis with node address A is 1 count.

Command Sent: #AMDB0001W; write the dead band setting.

Response Received: move to dead band

#AMDB0001W; Dead band settings for "PAN" axis with node address A is 1 count.

3) Motor Movement – Advanced

MLF (Motor Limit Forward): This command is used to configure the soft stop limit of the axis in the “clockwise direction”. The soft stop value are set in “decimal count”. For degree to count conversion refer to section “Position Calculation” and quick guide refer to “Degree to Count Conversion Table”. This command is “Read” and “Write” command.

Example:

Command Sent: **#AMLF0000R**; Read “Forward Limits” of “PAN” axis with node address A

Response Received: **read fwd limit**

#AMLF6023R; “Forward Limits” of “PAN” axis with node address A is 6023 count (+90 degrees)

Command Sent: **#AMLF6934W**; Set “Forward Limits” of “PAN” axis with node address A to 6934 count (+170 degrees)

Response Received: **set fwd limit**

#AMLF6934W; Acknowledgement of the command received from the “PAN” axis with node address A

MLB (Motor Limit Backward): This command is used to configure the soft stop limit of the axis in the “ counter clockwise direction”. The soft stop value are set in “decimal count”. For degree to count conversion refer to section “Position Calculation” and quick guide refer to “Degree to Count Conversion Table”. This command is “Read” and “Write” command.

Example:

Command Sent: **\$AMLB0000R**; Read “Backward Limits” of “TILT” axis with node address A

Response Received: **read back limit**

\$AMLB3976R; “Backward Limits” of “TILT” axis with node address A is 3976 count (-90 degrees)

Command Sent: **\$AMLB4488W**; Set “Backward Limits” of “TILT” axis with node address A to 4488 count (-45 degrees)

Response Received: **set back limit**

\$AMLB4488W; Acknowledgement of the command received from the “TILT” axis with node address A

MGF (Motor Go Forward): This command is similar to Move Motor Forward (MMF) and moves the motor in the “clockwise direction” at the speed specified in the data field value. The allowable value for data field is 0050 to 2000. The data field value 0050 means 0.5 degrees/sec and 2000 means 20.00 degrees/sec. Unlike MMF command, only one response is received from MGF command and the data field value of this response is the current position of the axis. This command is “Write” command only.

Example

Command Sent: **#AMGF1000W**; PAN axis with Node Address A move clockwise at speed of

10 degrees/sec (Data Field: 1000)

Response Received: **move forward**

#AMGF5513W; Acknowledgement received from the axis along with the current position of the PAN axis (Data Field: 5513 which is 45 degree)

MGB (Motor Go Backward): This command is similar to Move Motor Forward (MMB) and moves the motor in the “counter clockwise direction” at the speed specified in the data field value. The allowable value for data field is 0050 to 2000. The data field value 0050 means 0.5 degrees/sec and 2000 means 20.00 degrees/sec. Unlike MMB command, only one response is received from MGB command and the data field value of this response is the current position of the axis. This command is “Write” command only.

Example:

Command Sent: **\$BMGB1000W**; Move “TILT” axis with node address B in counter clockwise direction at speed of 10 degrees/sec (Data Field: 1000)

Response Received: **move backward**

\$BMGB5000W; Acknowledgement received from the “TILT” axis along with the current positional value of the axis (Data Field: 5000 which is 0 degree)

MHF (Motor Home Forward): This command moves the axis in “clockwise direction” to align the “zero marking” on output shaft with “index marking” of the incremental encoder and locate the actual “zero position” of the axis. This is called as “homing” of the axis. Each axis must be “homed” at the power-up; refer to section “Homing of SS250mkII/mkIII” for more details. The data field value is used to specify the desired speed for “Homing”. The allowable value for data field is 0050 to 2000. The data field value 0050 means 0.5 degrees/sec and 2000 means 20.00 degrees/sec. The axis will stop as soon as it reaches the “home” position. This command is “write” only. For this command two responses are received: first one to acknowledge that command is received and second indicates the current position of axis.

Example:

Command Sent: **#AMHF1000W**; Find the “Home” position of the “PAN” axis with node address A in clockwise direction at speed 10 degrees/sec

Response Received: **motor seek home forward**

#AMHF1000W; Acknowledgement of the command received from the “PAN” axis with node address A

#AMRL5001R; “PAN” axis with node address A is now “homed”
(i.e. 5000= 0 degree)

MHB (Motor Home Backward): This command moves the axis in “counter clockwise direction” to align the “zero marking” on output shaft with “index marking” of the incremental encoder and locate the actual “zero position” of the axis. This is called as “homing” of the axis. Each axis must be “homed” at the power-up; refer to section “Homing of SS250mkII/mkIII” for more details. The data field value is used to specify the desired speed for “Homing”. The allowable value for data field is 0050 to 2000. The data field value 0050 means 0.5 degrees/sec and 2000 means 20.00 degrees/sec. The axis will stop as soon as it reaches the “home” position. This command is “write” only. For this command two responses are received: first one to acknowledge that command is received and second indicates the current position of axis.

Example:

Command Sent: **\$BMHF0800W**; Find the “Home” position of the “TILT” axis with node address B in the counter clockwise direction at speed 8 degrees/sec

Response Received: **motor seek home forward**

\$BMHF1000W; Acknowledgement of the command received from the “TILT” axis with node address B

\$BMRL5001R; “TILT” axis with node address B is now “homed”
(i.e. 5000= 0 degree)

4) Diagnostics

TMP (Device Temperature): This command reads the “cpu” temperature in degrees celsius. This command is “Read” command only.

Example:

Command Sent: **\$ATMP0000R**; Read the temperature of “TILT” axis with node address A

Response Received: **cpu temperature**

\$ATMP0031R; Temperature of “TILT” axis with node address A is 31degC

5) Special Commands

GSA (Global Set Address): This Command is used to set the node address of all the Sidus Products connected to RS485 serial port such as PAN and TILTS, CAMERA, LASERS and LIGHTS. The ninth character of command defines the node address. The data range is A-Z or a-z, case sensitive. This command mnemonic is used along with “GLOBAL” Header (!) only. For this command not acknowledgement is generated. This command is “Write” command only

Example:

Command Sent: **!AGSA000BW**; Modify the node address “TILT” axis with node address A to B

GST (Global Stop): This single command is used to turn off all the Sidus Products connected to a RS485 serial port such as PAN and TILTS, CAMERA, LASERS and LIGHTS. This command mnemonic is used along with “GLOBAL” Header (!) only. The data bytes are ignored in this command. For this command not acknowledgement is generated. This command is “Write” command only.

Example:

Command Sent: **!BGST0000W**; Stops the functionality of all the devices with node address “B”

DBL (Device Boot Loader): This command used to update the axis with new firmware. Contact Sidus Solutions for more details.

MAL (Motor Actual Location) (SS250mkIV ONLY): This command sends the true position of the encoder. The value returned is the decimal value multiplied by 100. This command uses more value spacings than any other command to deliver the proper amount of decimal places. 6 value spacings are used rather than 4 on the received packet.

Example:

Command Sent: **#AMAL0000R**

Command Received: **#AMAL010235R** This would be 102.35 degrees.

Command Sent: **#AMAL0000R**

Command Received: **#AMAL-13245R** This would be -132.45 degrees

4.4.3. Configure MOXA Uport 1150 USB to Serial Bridge

Control of the SS250mkII/mkIII/mkIV are accomplished over a RS485/RS232 serial port. If the computer does not have a suitable serial port a USB to serial converter must be used. Sidus recommends using a MOXA UPort 1150 USB to serial Bridge.

To use the MOXA UPort 1150, or other similar devices, suitable drivers must be installed on the computer. Drivers are available for most operating systems and the most current version may be downloaded from the MOXA website:

https://www.moxa.com/support/sarch_result.aspx?type=soft&prod_id=380&type_id=9

To avoid malware, it is not recommended to download drivers from any other sites.

To install and configure the drivers for the MOXA UPort 1150 on a Windows7 OS (other Windows versions may vary slightly), do the following:

1. Download and Install the driver. At the date of writing, the most current version is **driv_win_uport1p_v1.9_build_16020210_whql.exe**
2. After installing the driver, open the “**Windows Device Manager**” to configure the driver
3. The MOXA Uport 1150 will be displayed as “**Multi-port serial adapters**”
4. Double-click on the “**Multi-port serial adapters**”

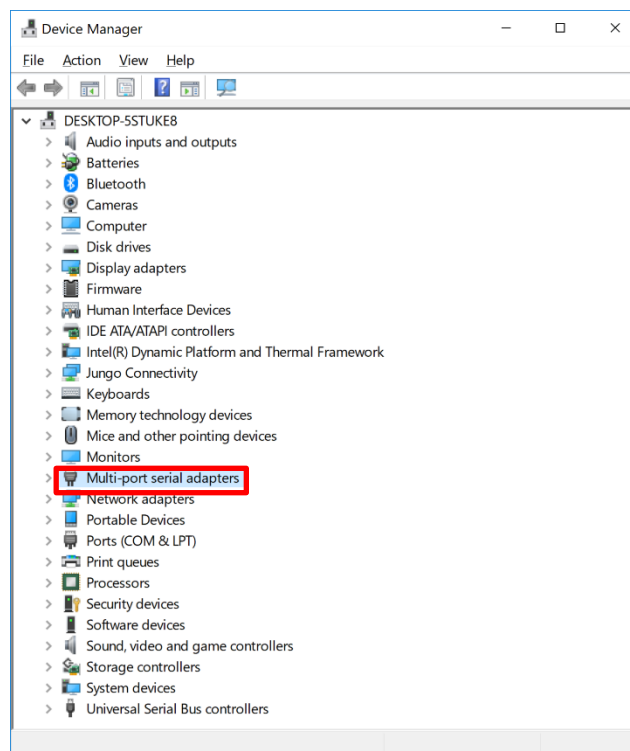


Fig.7: Multi Port Serial Adapter Shown in Windows Device Manager

5. This will display the **“UPort 1150 device”**

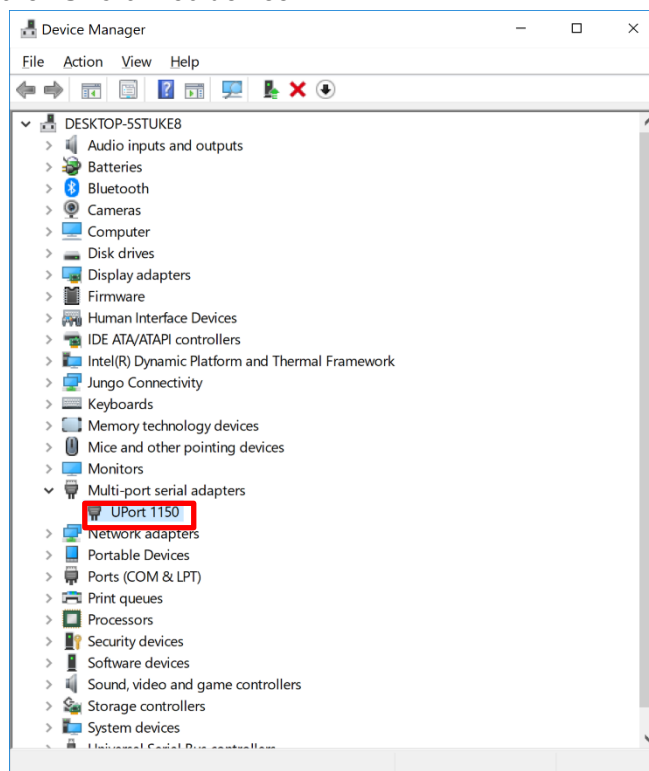


Fig.8: MOXA UPort 1150 Shown in Windows Device Manager

6. Double-click on the **“UPort 1150”**. This will display a pop-up window with the **“UPort 1150 Properties”**

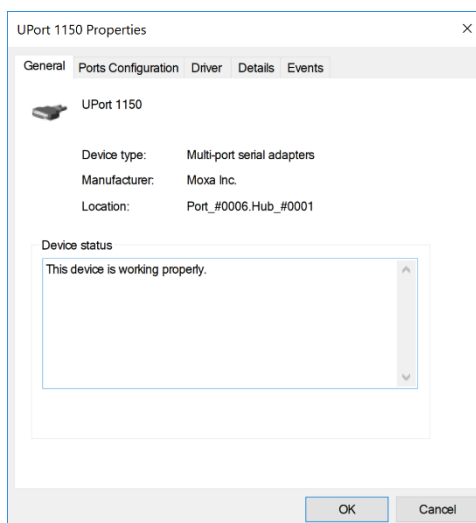


Fig.9: UPort 1150 Properties – General Tab

7. Click on the **“Port Configuration”** tab

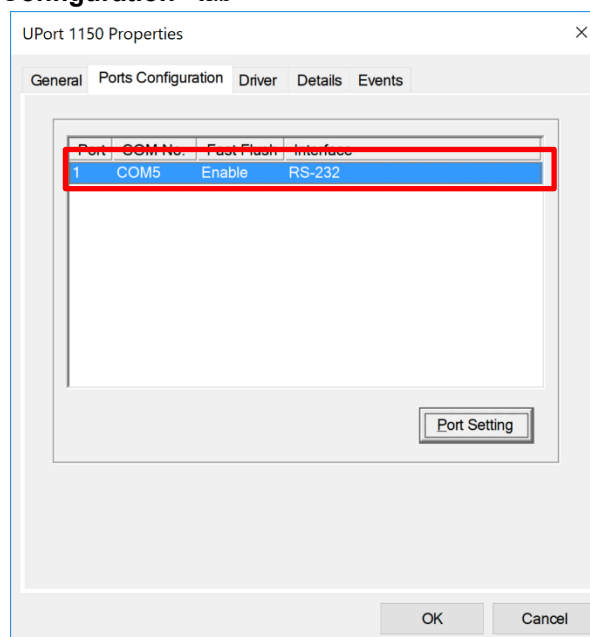
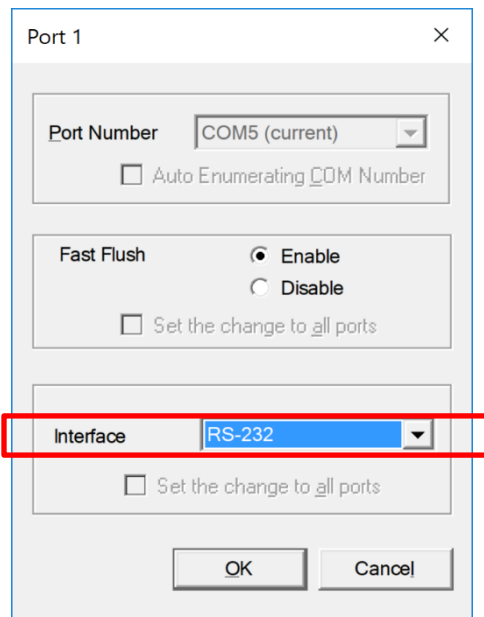


Fig.10: UPort 1150 Properties – Port Configuration Tab

8. The **“Port Configuration”** tab will display:
 - a. the com port number, in this example **“COM5:”**
 - b. the serial port interface, in this case **“RS-232”**
9. The **“Interface”** can now be changed to **“RS485-2W”** or **“RS232”** if required; by clicking on the **“Port Setting”** button to display the **“Port 1”** configuration window.



10. Click on the “**Interface**” drop-down box and select “**RS-485-2W**”

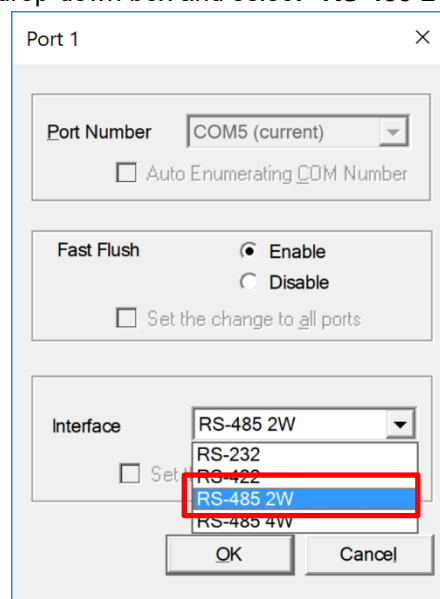


Fig.11: UPort 1150 Properties - Port 1 Configuration Window - Changing Interface Setting

11. Click on the “**OK**” button to set the port to RS-485-2W

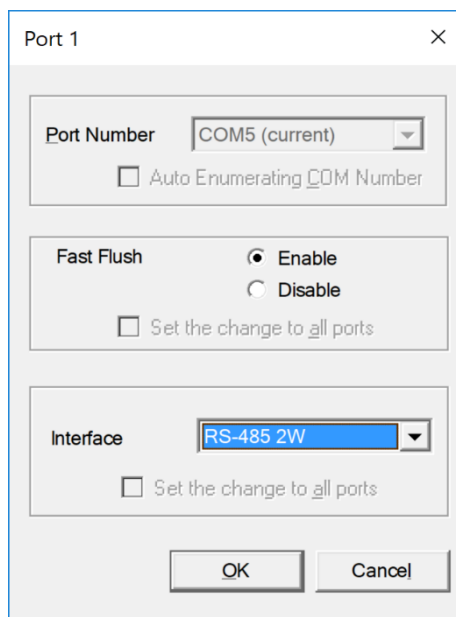


Fig12: UPort 1150 Properties - Port 1 Configuration Window - Interface Set to RS-485 2W

12. If needed, the **“Port Number”** can be changed at this time.
13. Click on the **“OK”** button to return to the **“UPort 1150 Properties”** window.
 - a.) Confirm that the **“Interface”** is set to **“RS-485-2W”** or as required
 - b.) Take note of the assigned **“Port Number”**, it will be required later on

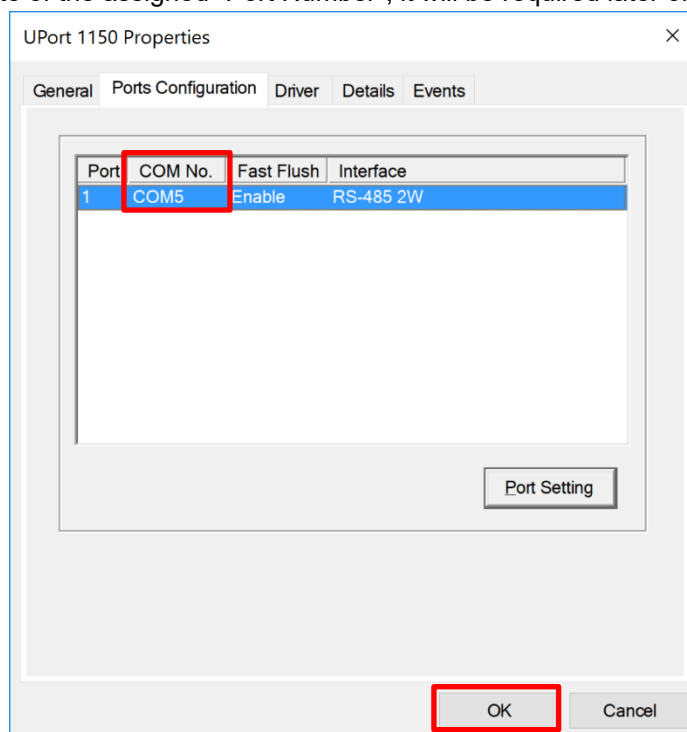


Fig.13: UPort 1150 Interface Set to RS-485 2W

14. Click the **“OK”** button to return to the **“Device Manager”** window

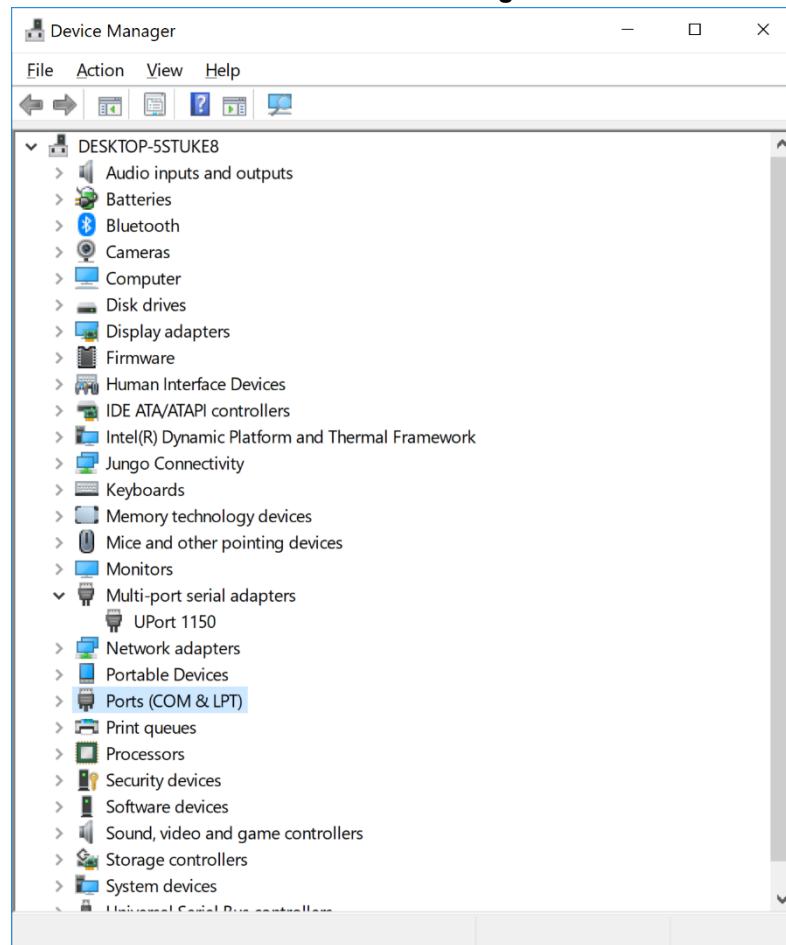


Fig.14: Windows Device Manager - Ports (COM & LPT)

15. In the **“Device Manager”** window, double-click **“Ports (COM & LPT)”** to display the available ports

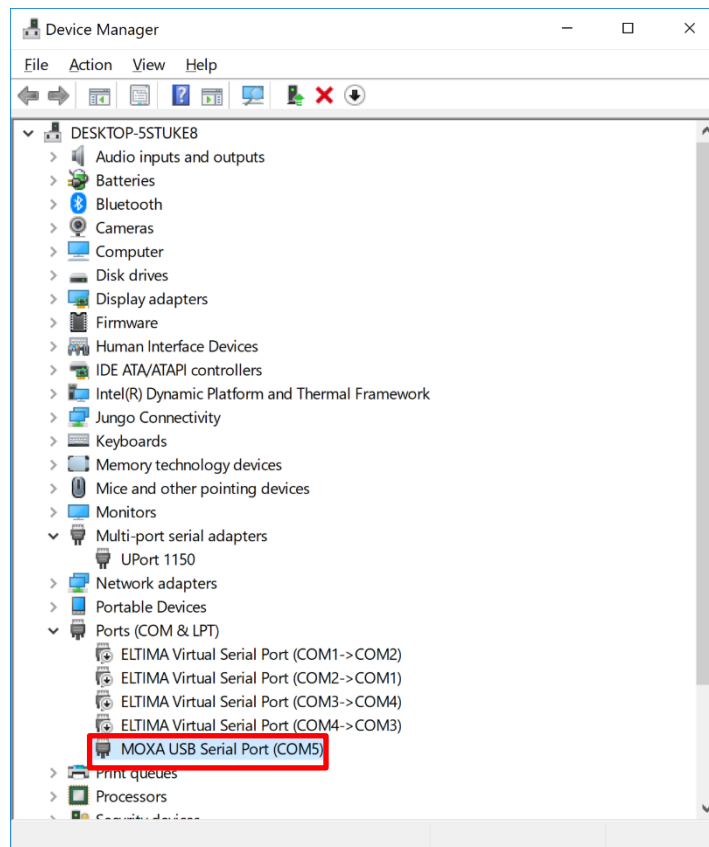


Fig. 15: Windows Device Manager Displaying Available COM Ports

16. Double-click the **“MOXA USB Serial Port (COM5)”** to open the **“MOXA USB Serial Port (COM5)”** Properties window
 - a.) Note: the COM port number will vary and may not always be COM5

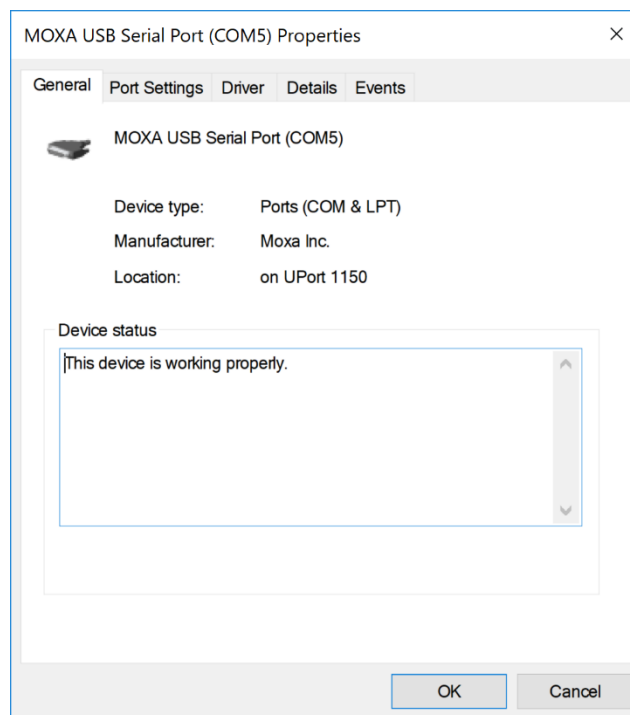


Fig.16: MOXA USB Serial Port (COM5) Properties – General Tab

17. Click on the “**Port Settings**” tab

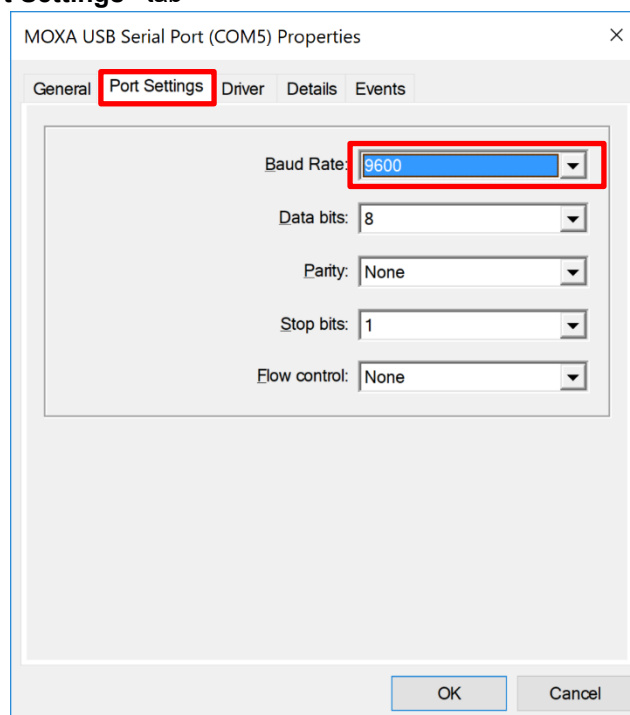


Fig.13: MOXA USB Serial Port (COM5) Properties – Port Settings Tab

18. Ensure the port settings are set as follows and as shown. If the factory default setting have been changed in the SS250mkII/ SS250mkIII/SS250mkIV, make the corresponding changes here:

UPort 1150 Port Settings	
Setting	Value
<u>B</u> aud Rate	9600
<u>D</u> ata Bits	8
<u>P</u> arity	None
<u>S</u> top Bits	1
<u>F</u> low Control	None

Table 2: UPort 1150 Port Settings Values

19. Click the “**OK**” button to return to the “**Device Manager**” window
20. This completes the installation and configuration of the MOXA UPort 1150 USB to serial bridge, the “**Device Manager**” may now be closed

4.4.4. Serial port Communication using Termite

The one of the method of communicating with the SS250mkII/mkIII/mkIV series are to use a simple terminal emulation program, such as “Termite”, running on a computer or laptop. Other more complicated terminal emulation programs, such as TeraTerm may also be used. Termite Version 3.4 is copyrighted software that is free for personal and commercial use. You may use it and distribute it without limitations. You may however not remove or conceal the copyright. There are no guarantees or warranties whatsoever; use it at your own risk. This program should only be downloaded from the CompuPhase website to ensure that it does not contain malware:

<https://www.compuphase.com/software/termite-3.4.exe>

This link will download a self-extracting setup file that contains Termite and all plug-ins and documentation (but excluding the source code for these plug-ins). The setup file can be used to copy the files to the hard disk. No installation is required but it may be convenient to manually create a shortcut on the desktop to the Termite.exe file or pin it to the Taskbar.

Because most laptops and computers no longer have a serial port, a USB to Serial adapter must be used. The *MOXA UPort 1150* USB to serial bridge is recommended for communicating with the SS250mkII/SS250mkIII. Sidus has found the UPort 1150 to be reliable and trouble-free in the field and it is also capable of communicating with either RS232 or RS485 interfaces. The UPort 1150 requires configuration as described in section VIII: Configure MOXA UPort 1150 USB to Serial Bridge.

To control the SS250mkII/mkIII/mkIV series over RS485/RS232 Serial Interface using Termite:

1. Do not turn power on to the SS250mkII/mkIII/mkIV until instructed to do so.
2. Open the Termite program, as shown in Fig. 14

- 2.1 In this window the serial port has been not yet been properly initialized

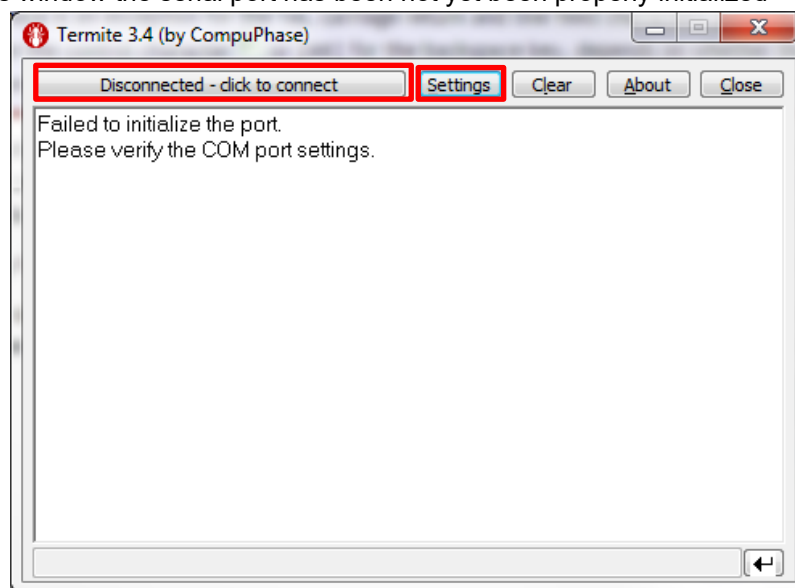


Fig. 14: Termite Power-Up Window – Port Not Connected

3. Click on the **“Settings”** button. This will open the “Serial port settings” popup window, as shown in *Fig. 15*.
 - 3.1 In the **“Port”** window, select the “Communications Port” assigned to the MOXA UPort 1150 by Windows. If the Communication Port is not known, it may be found using the **Windows Device Manager**. In this case **“COM1”** has been selected
 - 3.2 Ensure that the balance of the “Port configuration” settings are selected as shown in the figure
 - 3.2.1 These settings shown in the figure reflect the factory default settings. If the Baud rate or any other settings have been changed, make the appropriate changes
 - 3.3 In the **“Transmitted Text”** section only the following check boxes should be checked, no other:
 - 3.3.1 **“Append CR-LF”**
 - 3.3.2 **“Local echo”**
 - 3.4 In the “Options” section only the following check boxes should be checked, no others:
 - 3.4.1 **“Autocomplete edit line”**
 - 3.4.2 **“Keep history”**
 - 3.5 In the **“Plug-ins”** section no check boxes should be checked

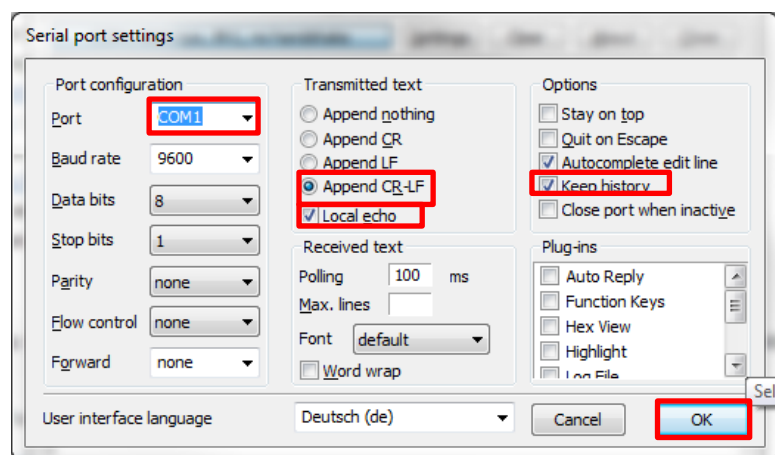


Fig.15: Termite Serial Port Settings Window

4. Click the “OK” button to accept the Serial port settings:
 - 4.1 The Termite window should open, as shown below
 - 4.2 “Termite is initialized and ready” should appear in the main window as shown below

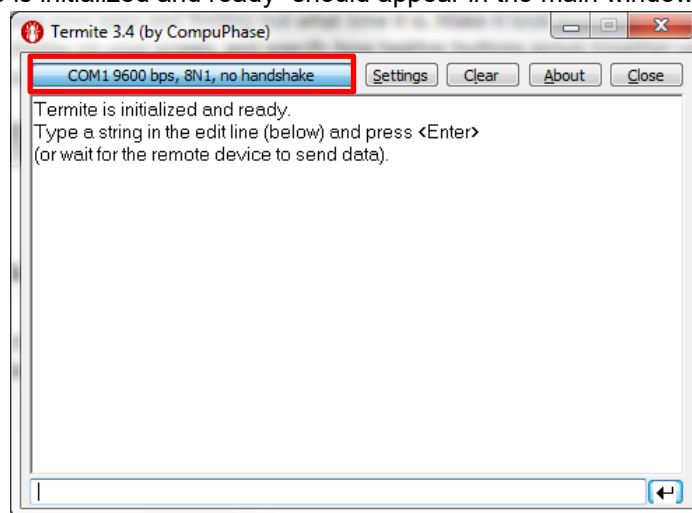


Fig.16: Termite Window, Port Recognized and Connected

5. Turn power on to the SS250mkII/ SS250mkIII/ SS250mkIV.
6. A valid power-up message should be displayed, as shown in Fig 3. In this figure the SS250mkII/mkIII/mkIV has a Node Address of “A”.

5. Analog Control – SS250 MkIII/SS250mkIV

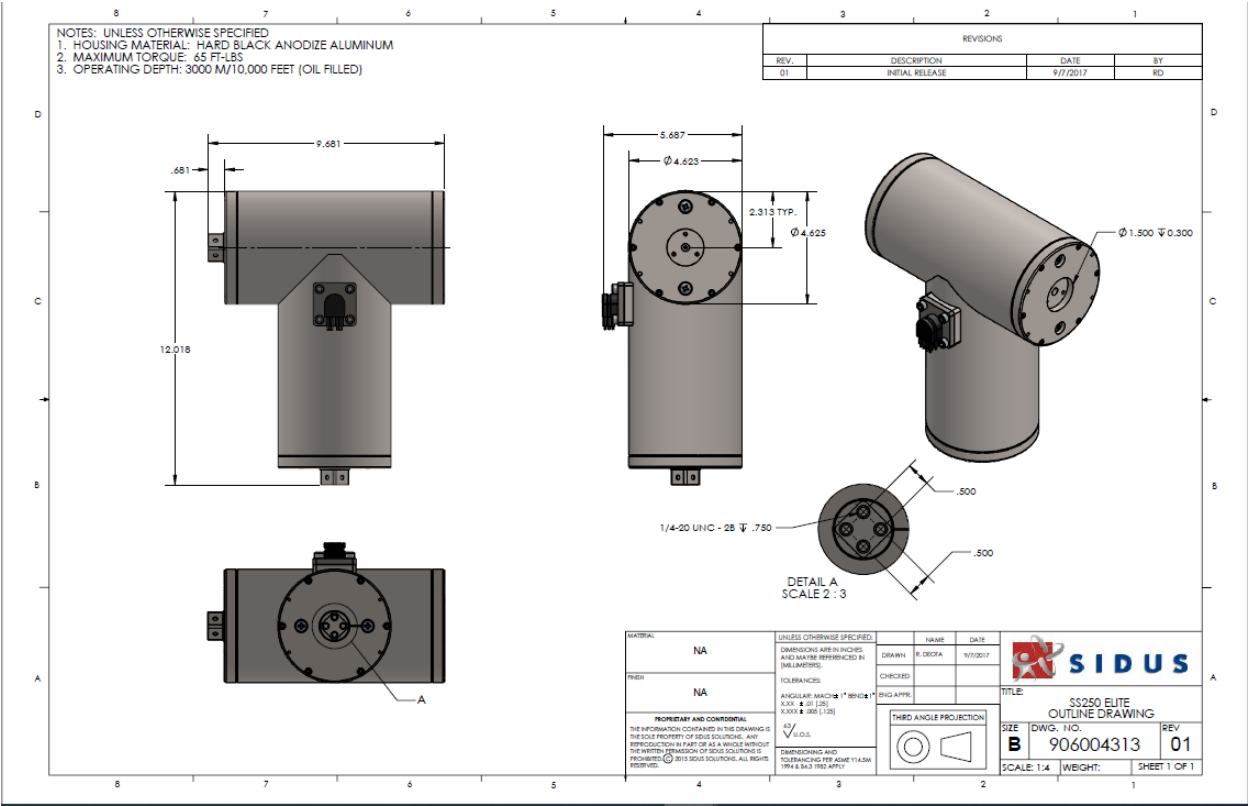
1. If the device is factory configured to be analog controlled, then Input voltages on the clockwise and counterclockwise pins determine the movement of the pan and tilt directions.
2. The recommended voltage for the P/CCW [Power counterclockwise] and P/CW [power clockwise] is 24 V DC. The analog voltages need to share the same ground as the power to the pan and tilt.
3. For information of analog pinout see section 2.3.1.
4. Motor position feedback for pan and tilt axis is also available on the 'FB Tilt' and 'FB Pan' pins. These feedback pins provide a voltage between 0 and 5V depending on the position. The following table gives the typical output voltage for a given angle.

Position (Degrees)	Voltage (Volts)	Position (Degrees)	Voltage (Volts)
-170	0.112	5	2.642
-165	0.252	15	2.781
-155	0.394	25	2.922
-145	0.531	35	3.062
-135	0.672	45	3.204
-125	0.812	55	3.344
-115	0.953	65	3.485
-105	1.094	75	3.628
-95	1.235	85	3.767
-85	1.375	95	3.906
-75	1.516	105	4.05
-65	1.657	115	4.2
-55	1.797	125	4.33
-45	1.938	135	4.48
-35	2.077	145	4.62
-25	2.218	155	4.77
-15	2.36	165	4.9
-5	2.501	170	4.94

Note: Pan and Tilt speed is fixed for the analog configuration and set during factory setup.

Outline Drawing- Reference Only

6. SS250mkII series /SS250mkIII/mkIV Drawing



Appendices

Appendix:1 Position Calculation

Calculating Position: Counts & Degrees

Typically, position is discussed in the unit of degrees. The values used for the commands are not in degrees but are based on the encoder within the unit. Encoder counts can be converted into degrees (and vice versa) using the following formula:

$$\text{Count Value} = \text{Degrees} / \text{Scale Factor of Encoder} + 5000$$

The encoder scale factor is **0.0879**.

For example, to find a position of 112 degrees:

$$112 / 0.0879 + 5000 = 6274 \text{ counts}$$

The command to move a pan motor at address A to 112 degrees is thus:

#AMML6274W

To find the degree position corresponding to 4297 counts:

$$(4297 - 5000) \times 0.0879 = -61.79 \text{ degrees}$$

The device feedback \$AMRL4297R thus means that the tilt motor axis is at -61.79 degrees.

NOTE:

For mkIV series devices, the angle is directly reported with the MAL command and the conversion above does not need to be applied. However, values returned from the MRL command still need the above conversion to be backwards compatible with systems designed for the mkII/mkIII series devices.

Appendix:2 Degrees to Encoder Count-Quick Reference Guide

Degree	Counts	Degree	Count	Degree	Count	Degree	Count
0	5000.000	195	7218.430	0	5000.000	-195	2781.570
5	5056.883	200	7275.313	-5	4943.117	-200	2724.687
10	5113.766	205	7332.196	-10	4886.234	-205	2667.804
15	5170.648	210	7389.078	-15	4829.352	-210	2610.922
20	5227.531	195	7218.430	-20	4772.469	-215	2554.039
25	5284.414	215	7445.961	-25	4715.586	-220	2497.156
30	5341.297	220	7502.844	-30	4658.703	-225	2440.273
35	5398.180	225	7559.727	-35	4601.820	-230	2383.390
40	5455.063	230	7616.610	-40	4544.937	-235	2326.507
45	5511.945	235	7673.493	-45	4488.055	-240	2269.625
50	5568.828	240	7730.375	-50	4431.172	-245	2212.742
55	5625.711	245	7787.258	-55	4374.289	-250	2155.859
60	5682.594	250	7844.141	-60	4317.406	-255	2098.976
65	5739.477	255	7901.024	-65	4260.523	-260	2042.093
70	5796.359	260	7957.907	-70	4203.641	-265	1985.210
75	5853.242	265	8014.790	-75	4146.758	-270	1928.328
80	5910.125	270	8071.672	-80	4089.875	-275	1871.445
85	5967.008	275	8128.555	-85	4032.992	-280	1814.562
90	6023.891	280	8185.438	-90	3976.109	-285	1757.679
95	6080.774	285	8242.321	-95	3919.226	-290	1700.796
100	6137.656	290	8299.204	-100	3862.344	-295	1643.914
105	6194.539	295	8356.086	-105	3805.461	-300	1587.031
110	6251.422	300	8412.969	-110	3748.578	-305	1530.148
115	6308.305	305	8469.852	-115	3691.695	-310	1473.265
120	6365.188	310	8526.735	-120	3634.812	-315	1416.382
125	6422.071	315	8583.618	-125	3577.929	-320	1359.499
130	6478.953	320	8640.501	-130	3521.047	-325	1302.617
135	6535.836	325	8697.383	-135	3464.164	-330	1245.734
140	6592.719	330	8754.266	-140	3407.281	-335	1188.851
145	6649.602	335	8811.149	-145	3350.398	-340	1131.968
150	6706.485	340	8868.032	-150	3293.515	-345	1075.085
155	6763.367	345	8924.915	-155	3236.633	-350	1018.203
160	6820.250	350	8981.797	-160	3179.750	-355	961.320
165	6877.133	355	9038.680	-165	3122.867	-360	904.437
170	6934.016	360	9095.563	-170	3065.984	-215	2554.039
175	6990.899	215	7445.961	-175	3009.101	-220	2497.156
180	7047.782	220	7502.844	-180	2952.218	-225	2440.273
185	7104.664	225	7559.727	-185	2895.336	-230	2383.390
190	7161.547			-190	2838.453		

Appendix:3 Homing Calibration Routine of SS250mkII/mkIII series

Problem Description:

The current design of “SS250mkII/mkIII series” (henceforth referred to as “unit”) uses incremental encoder on both axes to accurately measure the positional feedback value. One of the traditional limitation of the incremental encoder is that it may lose its position when powered down and requires “homing” routine at power up to realign its encoders index to “actual zero degree”.

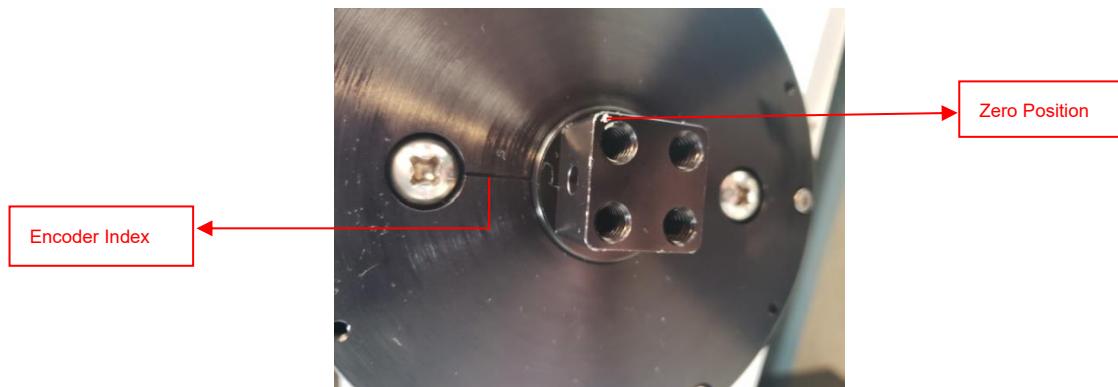
Because of the incremental encoders, both the axis of SS250mkII/mkIII losses its last positional feedback value/count when the unit is powered down. In addition to this, every time the unit is powered up, axes reset its positional feedback value/count to zero degree (i.e. 5000 count value). These results in the following problems:

- 1) Every time after power up, if the unit is not “Homed” then positional feedback value/count provided by the axis is inaccurate.
- 2) There are chances that the axis might “hit” installed mechanical “Hard Stop”.

Hence, it is very important to perform “Homing” routine/ procedure every time when the unit is powered up.

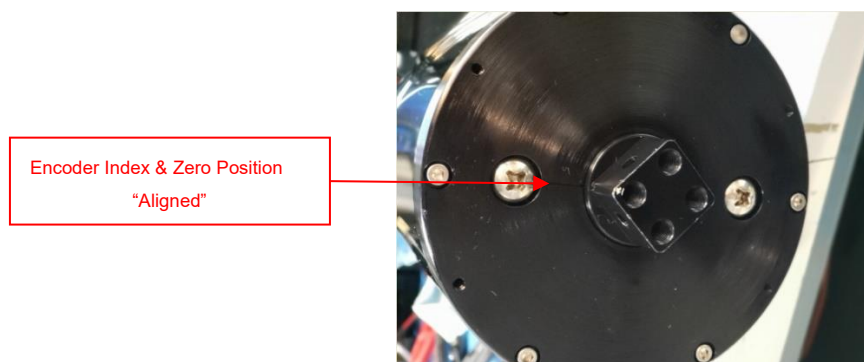
Solution:

Indication of “Home” Position: On each axis, the actual zero degree position is at the center of the available range of the motion. To identify actual “zero degree”, there are two marking available; one on the encap (henceforth referred to as “encoder index”) and other on the output shaft (henceforth referred to as “zero position”) as shown below:



The mechanical “Hard Stop” installed and count/value set for “Soft Stop” limits the movement of the axis in both the direction (i.e. Clockwise and Counter Clockwise direction).

When “encoder index” marking on the encap is aligned with “zero position” marking on the output shaft; the axis is said to be at the “home” position (i.e. axis is at actual zero degree) as show below:



And when both the axes are at the “home” position, the unit is said to be “homed”.

Sidus Solution Command used for “Homing”: There are two “Sidus Solutions Protocol” commands used for “Homing” the axes. Each command is explained below with example:

- 1) Motor Home Forward (MHF): Every time “SS250mkII/mkIII series” products is powered up; it must be “homed” (i.e. align the “zero position” marking on the output shaft with “encoder index” marking on the encap for both the axes). This command is used to “Home” the axis in the **clockwise direction** at speed specified in the data field. . The allowable data field value is 0010 (i.e. 0.10 degree/sec) to 2000 (i.e. 20.00 degree/sec). The output shaft of the axis will stop as soon as it detect “encoder index” marking on the “encap”.

Example:

Command Sent: **#AMHF1000W**; “Home” PAN Axis with Node address: A @ 10 deg/sec.

Response Received: **motor seek home forward**

#AMHF1000W; Acknowledgement Received

#AMRL5001R; PAN Axis with Node address: A “is now” “Homed”

(i.e. feedback count/value=5001 which zero degree)

- 2) Motor Home Backward (MHB): Every time “SS250mkII/mkIII series” products is powered up; it must be “homed” (i.e. align the “zero position” marking on the output shaft with “encoder index” marking on the encap for both the axes). This command is used to “Home” the axis in the **counter clockwise direction** at speed specified in the data field. . The allowable data field value is 0010 (i.e. 0.10 degree/sec) to 2000 (i.e. 20.00 degree/sec). The output shaft of the axis will stop as soon as it detect “encoder index” marking on the “encap”.

Example:

Command Sent: **#BMHF1200W**; “Home” TILT Axis with Node address: B @ 12 deg/sec.

Response Received: **motor seek home backward**

#BMHB1200W; Acknowledgement Received

#BMRL5001R; TILT Axis with Node address: B “is now” “Homed”

(i.e. feedback count/value=5000 which zero degree)

Homing Routine/Procedure: These “Homing” routine/procedure can be execute for any of the following situations:

- (1) When unit is out of visibility range.
- (2) When the unit is installed as a part of the application.
- (3) When the unit is being tested in standalone condition on the workbench.

Steps to execute “Homing” routine/procedure are:

- (1) PAN Axis “Homing” in Clockwise Direction: First send command **#AMHF1000W** (For “Homing” recommended speed is 10degrees/sec). And check the response.

If Response received is: **motor seek home forward**

#AMHF1000W

#AMRL5001R

➔ Don’t Execute Step (2). PAN Axis is “Homed”

If Response received is: **motor seek home forward**

#AMHF1000W

motor slipped

#AMRL5915W

➔ Execute Step (2); as PAN Axis needs to be “Homed” in counter clockwise direction

- (2) PAN Axis “Homing” in Counter Clockwise Direction: Send command “#AMHB1000W” (For “Homing” recommended speed is 10degrees/sec). And check the response.

If Response received is: motor seek home forward

#AMHB1000W

#AMRL5001R

➔ PAN Axis is now “Homed”

If Response received is: motor seek home forward

#AMHB1000W

motor slipped

#AMRL4886W

➔ Contact Sidus Solutions for further assistance.

It is advisable to “Home” one axis at a time, but not necessary. Also, the order of axis is also inconsequential.

Appendix:4 Homing Calibration Routine of SS250mkIV series

The SS250mkIV does not need to undergo the homing routine as the mkII and mkIII series does. This is accomplished through a redesigned encoder system. An absolute encoder is used in place of an incremental encoder as the mkII/mkIII used. However, the commands for the homing sequence are still supported and will return the device to its zero position. Rezeroing the device in the field is available, contact Sidus support if the zero location does not seem to align with the zero notches on the output shaft and end cap.

Appendix:5 Noise Reduction on SS250mkIII series

Problem Description:

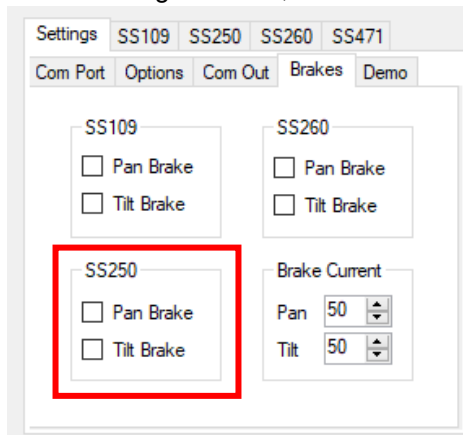
The SS250mkIII series currently features a braking system that enables the unit to be magnetically locked in place after movement. This locking mechanism is particularly useful when the unit is transporting heavy loads. However, the braking system occasionally results in audible noise that may affect systems that require minimal to no noise.

Solution:

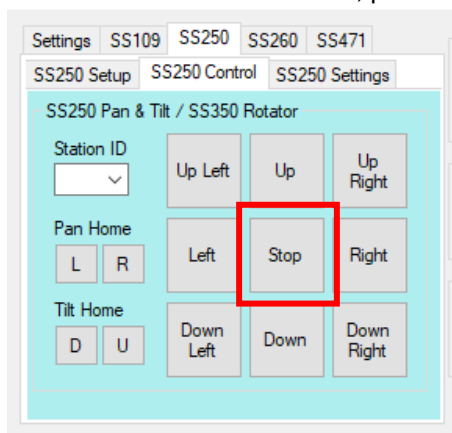
Turning off Brakes: Users have the option to disable the brakes, which reduces noise levels. However, this adjustment may pose challenges for applications dealing with heavier loads.

Turning off Brakes using the Sidus GUI:

- 1) Under Settings>Brakes, uncheck the Tilt and Pan



- 2) Under SS250>SS250 Control, press stop to disable the brakes.

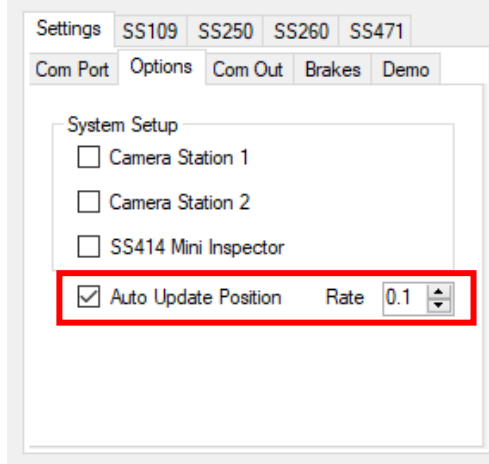


Turning off Brakes using a Terminal Emulator (Termite):

- 1) Send the command "#AMBS0000W" to disable PAN brake.
- 2) Send the command "\$AMBS0000W" to disable TILT brake.
- 3) Send the command "#AMST0000W" to stop PAN motor and turn off brake.
- 4) Send the command "\$AMST0000W" to stop TILT motor and turn off brake.

Higher Frequency Interference (GUI users only): If the device is still causing interference and you are using the GUI, the noise could be caused by the 'Auto Update Position' feature.

- 1) Under Settings>Options there is a checkbox that says 'Auto Update Position'. Unchecking this will stop comm activity until another button is pressed.



Additional Solution (GUI users only): Sometimes the GUI will display the brakes as off when they are on. The easiest way to check whether the brakes are on is to see how much current is being drawn when it is in an idle state. If the brakes are engaged, the device should be pulling 500mA or more. If the brakes are not engaged, they should pull less than 100mA.

Customer Notes
